

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

B.Tech. EE Sem - V Examination, 2011-12
EE - 302 Electrical Power System -II

Date: 15/12/2011, Thursday

Time: 10 AM TO 1:00 PM

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

SECTION - I

- Q-1 A. Why the shapes of pin insulators have petticoat shape? 3
- B. An over head line is supported on two poles 200 meters apart having a difference in level of 10 meters. The conductor diameter is 2.0 cm and weight 2.3 Kg per meter length. Calculate the sag at the lower support under the conditions if wind provides a pressure of 57.5 Kg/m² of projected area and a factor of safety is 4. The maximum tensile strength of conductor is 4220 Kg/cm² 4
- Q-2 A. Explain the selection & spacing of conductor for EHV-AC transmission line 5
- B. Explain the term "Critical corona voltage" with reference to corona. 2
- C. A 3-phase, 220 KV, 50 Hz transmission line of length 200 Km consist of 1.5 cm radius conductor spaced 2 meters apart in equilateral triangular formation. If the temperature is 40°C and atmospheric pressure is 76 cm, calculate corona loss with help of Peek's formula. Take $m_0=0.85$ 4
- D. Discuss the points to be considered during the selection of voltage of transmission line. 3
- OR
- Q-2 A. Write technical note on "Radio Interference" 7
- B. As a design engineer if you are ask to decrease the corona loss at a design stage, which means you follow? Why? 2
- C. A certain 3-phase equilateral transmission line has a total corona loss of 53 KW at 106 KV and a loss of 98 KW at 110.9 KV. What is the disruptive critical voltage between lines? What is the corona loss at 113 KV? 5
- Q-3 A. How the ground wire protects the transmission line from induced over voltages from charged clouds & direct lightning strokes? 3
- B. Explain the vibrations of transmission line conductor, its effect and its mitigation technique. 4
- C. Write technical note on "Insulation Coordination" 7

OR

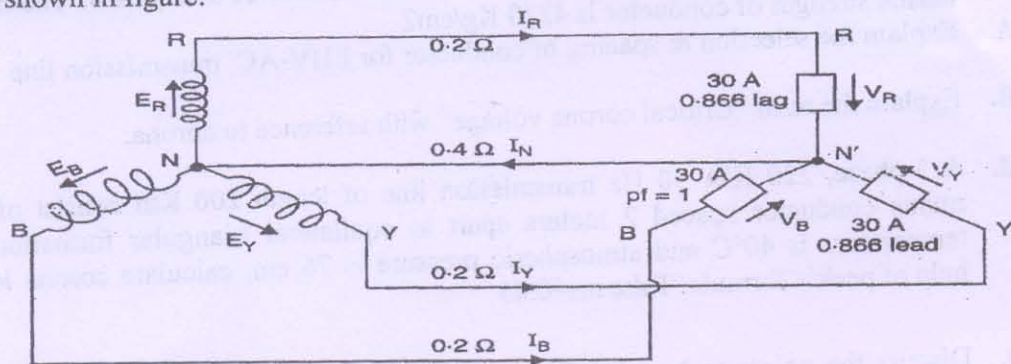
- Q-3 A. List out main equipments of substation & explain the function of each. 6
- B. List out the advantages of interconnection. 2
- C. Draw well labelled bus bar arrangement of 6
- (i) Double bus bar scheme
- (ii) One and a half breaker scheme.

SECTION - II

- Q-4 (A) Explain the phenomenon of Lamp Flicker with its classification and also suggest any Two Remedies to solve Lamp Flicker Problems. 4

- (B) What are the Practical limitations of Kelvin's Law? 3

- Q-5 (A) A 3- Φ , 4-wire distributor supplies a balanced voltage of 400/230 V to a load consisting of 30 A at p.f. 0.866 lagging for R-phase, 30 A at p.f. 0.866 leading for Y phase and 30 A at unity p.f. for B phase. The resistance of each line conductor is 0.2Ω . The area of X-section of neutral is half of any line conductor. Calculate the supply end voltage for R phase. The phase sequence is RYB and equivalent circuit diagram is shown in figure. 7



- (B) A 2-wire D.C. distributor AB is fed from both ends. At feeding point A, the voltage is maintained as at 230 V and at B 235 V. The total length of the distributor is 200 metres and loads are tapped off as under : 7

25 A at 50 metres from A ; 50 A at 75 metres from A
30 A at 100 metres from A ; 40 A at 150 metres from A.

The resistance per kilometre of one conductor is 0.3Ω

Calculate :

- (i) currents in various sections of the distributor
(ii) Minimum voltage and the point at which it occurs.

OR

- Q-5 (A) Explain types of Distribution System Arrangements & also draw neat well labeled diagram of these systems. 8

- (B) Write a Technical Note on Voltage Regulation and Explain any four methods of voltage regulation used on distribution systems in detail. 6

Q -6 (A) How does A.C. Distribution differ from D.C. distribution systems? Also state the advantages of AC distribution systems. 4

(B) A 2-conductor cable 1 km long is required to supply a constant current of 200 A throughout the year. The cost of cable including installation is Rs. $(20a + 20)$ per metre where 'a' is the area of X-section of the conductor in cm^2 . The cost of energy is 5P per kWh and interest and depreciation charges amount to 10%. Calculate the most economical conductor size. Assume that resistivity of conductor material to be $1.73 \mu\Omega \text{ cm}$. and length of line is 10^5 cm . 4

(C) Explain the advantages and limitations of Single Wire Earth Return (SWER) in connection with remote rural electrification. 6

OR

Q -6 (A) What is meant by component reliability and power system reliability? How is the Power system reliability measured? 4

(B) A distribution system has the following loads 4

8 a.m. to 11 p.m.	1.3 MW at 0.75 lagging p.f.
11 p.m. to 8 a.m.	0.4 MW at 0.8 lagging p.f.

Find the KVAR rating of fixed and switched capacitors to improve the p.f. to 0.95 lagging during the whole day.

(C) List out the common causes of unsatisfactory performance of rural distribution system and also suggest the remedial measures for the same. 6
